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Scientific interests: computational hydrodynamics; supernovae; supernova remnants; radiative processes; stellar evolution

(a) Research experience

ASIAA Taiwan	Distinguished Postdoctoral Fellow	September 2026 - present
University of Virginia	VITA Postdoctoral Research Fellow	September 2024 - August 2026
Purdue University	Visiting Researcher	May 2024 - August 2024

(b) Education

Purdue University	Physics and Astronomy	Ph.D.	January 2021 - August 2024
Purdue University	Physics and Astronomy	M.S.	August 2018 - August 2020
Jadavpur University	Electrical Engineering	B.E.	August 2014 - June 2018

(c) Fellowships and Awards

Distinguished Postdoctoral Fellowship, ASIAA Taiwan, 2026
KVPY (Kishore Vigyani Protsaahan Yojana) Fellowship, awarded by the Govt. of India, 2014
IAU Travel Grant towards the attendance of IAU Symposium 406, 2026

(d) Externally funded grants

Co-I for a NASA JWST grant on tracing ejecta and dust evolution in the Type Ia Supernova Remnant 0509-67.5, with a total observing time of 40.5 hours.

(e) Publications

(i) *First author*

1. **Soham Mandal**, Raphael Baer-way, Shazrene Mohamed, Irene Salmaso, Poonam Chandra, and Maryam Modjaz, Same Explosion, Many Faces: Numerical Modeling Reveals Viewing Angle as a Driver of Diversity for Core-collapse SNe in Binary Systems, arXiv e-prints , arXiv:2607.28519 (2026), arXiv:2607.28519 [astro-ph.HE] .
2. **Soham Mandal**, Parviz Ghavamian, Priyam Das, Ivo Rolf Seitenzahl, Shazrene Mohamed, and Ashley J. Ruiter, Deciphering the Explosion Mechanism of Type Ia Supernovae Using Their Remnants. II. A Deep Dive into Double Detonations with SNR 0509-67.5, The Astrophysical Journal **1002**, 155 (2026), arXiv:2509.02422 [astro-ph.HE] .
3. **Soham Mandal**, Núria Torres-Albà, Carles Badenes, and Shazrene Mohamed, Deciphering the Explosion Mechanism of Type Ia SNe Using Their Remnants. I. General Properties and a Case Study on Tycho's Remnant, The Astrophysical Journal **1004**, 53 (2026), arXiv:2508.10752 [astro-ph.HE] .
4. **Soham Mandal**, Paul C. Duffell, Abigail Polin, and Dan Milisavljevic, Measurement of Anisotropies in Supernova Remnant Observations and Their Interpretation Using Numerical Models, The Astrophysical Journal **972**, 87 (2024), arXiv:2403.12264 [astro-ph.HE] .

5. **Soham Mandal**, Paul C. Duffell, Abigail Polin, and Dan Milisavljevic, A 3D Numerical Study of Anisotropies in Supernova Remnants, The Astrophysical Journal **956**, 130 (2023).
6. **Soham Mandal** and Paul C. Duffell, Sprout: A moving-mesh hydro code using a uniformly expanding cartesian grid, The Astrophysical Journal Supplement Series **269**, 30 (2023).
7. **Soham Mandal**, Paul C. Duffell, and Yuan Li, Numerical Investigation of Dynamical and Morphological Trends in Relativistic Jets, The Astrophysical Journal **935**, 42 (2022).
8. **Soham Mandal** and K. K. Mandal, Optimal energy management of microgrids under environmental constraints using chaos enhanced differential evolution, Renewable Energy Focus **34**, 129–141 (2020).

(ii) *Co-author*

1. Z. Ding, D. Milisavljevic, T. Martin, T. Temim, J. C. Raymond, **S. Mandal**, and L. Drissen, 3D Kinematic Reconstruction of the Crab Nebula that Includes the Northern Ejecta "Jet", The Astrophysical Journal **1007**, 130 (2026).
2. T. Martin, D. Milisavljevic, T. Temim, **S. Mandal**, P. Duffell, L. Drissen, and Z. Ding, The non-uniform expansion of the Crab Nebula, Monthly Notices of the Royal Astronomial Society **544**, 1–12 (2025), arXiv:2502.19632 [astro-ph.HE] .
3. Miranda Pikus, Paul Duffell, Soham Mandal, and Abigail Polin, A Novel Parameterization for Rapid Cooling in Supernova Remnants, with Applications to the Pa 30 Nebula, The Astrophysical Journal **1002**, 104 (2026), arXiv:2511.08519 [astro-ph.HE] .
4. Paul C. Duffell, Abigail Polin, and **Soham Mandal**, Sculpting the Morphology of Supernova Remnant Pa 30 via Efficient Ejecta Cooling, arXiv e-prints , arXiv:2403.13641 (2024), arXiv:2403.13641 [astro-ph.HE] .
5. M. L. Lister, D. C. Homan, Y. Y. Kovalev, **S. Mandal**, A. B. Pushkarev, and A. Siemiginowska, TXS 0128+554: A Young Gamma-Ray-emitting Active Galactic Nucleus with Episodic Jet Activity, The Astrophysical Journal **899**, 141 (2020).
6. K.K. Mandal, **S. Mandal**, B. Bhattacharya, and N. Chakraborty, Non-convex emission constrained economic dispatch using a new self-adaptive particle swarm optimization technique, Applied Soft Computing **28**, 188–195 (2015).

(f) Invited talks

Institute of Astronomy and Astrophysics, Academia Sinica, Colloquium	*January 2026*
California Institute of Technology, TAPIR seminar	*December 2025*
Columbia University, THEA seminar	*October 2025*
Center for Computational Astrophysics, Stars and Plasma Astrophysics meeting	*October 2025*
Center for Astrophysics, ITC luncheon seminar	*September 2025*
California Institute of Technology, TAPIR seminar	*January 2024*
CIERA, Northwestern University, special seminar	*October 2023*
New York University, special seminar	*October 2023*
Center for Computational Astrophysics, compact objects meeting	*October 2023*
Columbia University, THEA seminar	*October 2023*
Carnegie Observatories, lunch seminar	*September 2023*
UC Berkeley, TAC seminar	*September 2023*
Tata Institute for Fundamental Research, State Of The Universe seminar	*June 2023*
Carnegie Observatories, tea talk	*May 2022*

(g) Contributed talks/posters

Rise_Time, Purdue University, USA	*August 2026 (Talk)*
IAU Symposium 406, Turku, Finland	*May 2026 (Talk)*
SN100 Conference, Stockholm, Sweden	*Aug 2025 (Poster)*
Rutgers Summer Transient Soiree, Rutgers University, USA	*July 2025 (Talk)*
Annual Symposium, National Radio Astronomy Observatory, USA	*May 2025 (Talk)*
Rise_Time, Purdue University, USA	*August 2024 (Talk)*
Supernova Remnants III, Crete, Greece	*June 2024 (Posters)*
Supernova Explosions (SNEX) Conference, Technion University, Israel	*September 2023 (Talk)*
Michael Faraday IET International Summit, Kolkata, India	*September 2015 (Talk)*

(h) Teaching

Summer Lecturer, Purdue University, PHYS 221 (Electrodynamics)	*Summer 2019, 2020*
Teaching Assistant, Purdue University, PHYS 536 (Modern electronics lab)	*Spring 2021*
Teaching Assistant, Purdue University, PHYS 172 (Modern mechanics)	*Spring and Fall 2019*
Teaching Assistant, Purdue University, PHYS 263 (Intro to astronomy)	*Fall 2018*

(i) Skills

Programming Languages	C/C++, Python, MATLAB
Software & Tools	MPI, OpenMP, OpenACC, PyTorch, HTML, LaTeX

(j) Synergistic activities

1. **Speaker, Astronomy on Tap, Charlottesville** *Fall 2025*
2. **LOC Member, Kavli Summer Program in Astrophysics 2025** *Summer 2025*
   (Hosted by the University of Virginia)
3. **LOC Member, Rise_Time 2024, Purdue University** *August 2024*
4. **Guest Speaker, The Astrophysics Podcast** *June 2024*
   (Link: https://rss.com/podcasts/astrophysics/1494506)
5. **Mentor, Inside Out program, Purdue University** *Summer 2022, 2023*
   Led a hands-on activity session for high school students where they designed and built a scaled down model of the solar system.
6. **Vice President, Physics Graduate Students Association, Purdue University** *2021 - 2022*
   Organized events to enhance interaction in the grad students community at Purdue.
7. **Event lead, Circuistic, Jadavpur University** *Fall 2017*
   Designed innovative problem statements for an analog electronic circuit building competition aimed at undergraduates at Convolution, the annual tech fest at Jadavpur University.